

AHANAF MOKAMMEL OMI

Full-Stack Developer

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Summary

Junior Full-Stack Developer with experience in ASP.NET MVC, C#, SQL Server, JavaScript, HTML5, and CSS3. Skilled in developing end-to-end full-stack web applications, including e-commerce platforms, database-driven systems, distributed job processing, incident response, operational risk simulation, and AI/ML applications. Strong problem-solving, collaboration, and Agile development expertise.

Experience

Junior Web Developer – Billing Master Software Ltd.

2025 – Present

Dhaka, Bangladesh

- Built 3+ ASP.NET MVC modules, improving internal processing efficiency by 30%.
- Implemented scalable CRUD and database features that reduced user-reported errors by 25%.
- Optimized SQL queries resulting in 20% faster data retrieval.

Executive Member – BRAC University Computer Club

2022 – 2025

Dhaka, Bangladesh

- Organized university-wide coding workshops and hackathons with 100+ student participants.
- Collaborated with the web team to maintain club website and event registration systems.

General Member – Dhaka Residential Model College Computer Club

2018 – 2020

Dhaka, Bangladesh

- Participated in intra-school programming competitions and basic web development training sessions.

Projects

NEXUS – Supply Chain Intelligence Command Center

[GitHub](#)

Technologies: ASP.NET Core MVC, C#, PostgreSQL, Redis, EF Core, Docker

- Engineered an enterprise supply-chain decision-support platform modeling suppliers, components, factories, warehouses, products, and customers as dependency graphs; implemented BFS/DFS traversal, alternate-path discovery, single-point-of-failure detection, bottleneck analysis, and downstream impact propagation.
- Built deterministic day-by-day disruption simulation with inventory depletion, stockout-day, service-level, recovery-time, revenue-at-risk and composite-risk calculations, multi-scenario/what-if analysis, mitigation heuristics, contract retrieval, natural-language scenario parsing, and evidence-first AI specialist orchestration; added RBAC, multi-tenancy, audit logging, SignalR, durable background jobs, OpenTelemetry and Testcontainers.

VERITAS – Enterprise Identity & Access Governance Platform

[GitHub](#)

Technologies: ASP.NET Core, C#, PostgreSQL, Redis, EF Core, Identity

- Built an enterprise IAM and zero-trust authorization platform supporting RBAC, ABAC, deny-by-default policy evaluation, ALLOW/DENY/REQUIRE-APPROVAL/REQUIRE-MFA outcomes, immutable policy versions, reproducible authorization decisions, risk scoring, privileged access and separation-of-duties controls.
- Implemented risk-aware MFA/approval workflows using IP, failed-login, privileged-action, production, after-hours, device-trust and location signals; added JIT/temporary access, access reviews, API-key hashing/rotation/revocation, HMAC-SHA256 webhooks and comprehensive audit trails.

NEURA – Neural AI Agent Orchestration & Continuity Platform

[GitHub](#)

Technologies: ASP.NET Core, C#, PostgreSQL, Redis, SignalR, OpenAI, Anthropic, Google AI

- Developed a multi-agent orchestration platform that maintains long-running task continuity across AI providers through context monitoring, structured ContextHandoffPackage generation, validation, automatic handoff, explainable weighted agent routing, persistent missions and provider abstraction.
- Implemented memory, knowledge graphs, real AI adapters for OpenAI/Anthropic/Google, retry/backoff, circuit breakers, rate limiting, encrypted provider credentials, cost/token tracking, persisted execution/audit records and real-time SignalR/Cytoscape mission visualization with E2E testing.

FORGE – AI Software Engineering Intelligence Platform

[GitHub](#)

Technologies: ASP.NET Core, C#, Roslyn, PostgreSQL, EF Core, Git

- Engineered a software-engineering intelligence pipeline combining Roslyn semantic/compiler analysis, repository in-

dexing, C# symbol extraction, dependency graphs, Git-history mining and deterministic security, architecture and technical-debt analysis.

- Implemented deterministic software-risk scoring, security pattern scanning, namespace architecture policies, repository/context retrieval, multi-tenant Identity/authZ, audit logging, health checks, rate limiting, unit/integration/architecture tests and modular-boundary enforcement.

Pulse – Distributed Job Processing & Workflow Platform

[GitHub](#)

Technologies: ASP.NET Core MVC, C#, RabbitMQ, Redis, SQL Server, EF Core

- Engineered a distributed job-processing modular monolith using RabbitMQ durable priority queues, manual acknowledgements, prefetch control, dead-letter exchanges and queues, with Redis distributed locks, idempotency protection and worker heartbeat management.
- Implemented Pending-to-Completed job lifecycle, failure/retry transitions, configurable exponential backoff, cancellation, scheduled jobs, worker orchestration, queue/worker monitoring, failure simulation and role-based operational management.

BD Price Finder – Multi-Vendor E-Commerce Platform

[Website](#)

Technologies: ASP.NET MVC, Razor Views, C#, HTML, CSS, JavaScript, SQL Server

- Developed a full-stack Bangladesh multi-vendor price-comparison/e-commerce marketplace supporting buyers, sellers, vendors and corporate stakeholders through role-specific dashboards, product/vendor workflows, approval processes and hierarchical permissions.
- Implemented product lifecycle management, dynamic variants, inventory/order management, 2-step email OTP authentication, coupons, EMI, warranty, services, accessories, Elasticsearch search, Redis caching, responsive Razor interfaces and vendor/location-aware marketplace functionality.

Local Agent Platform – Local-First Autonomous Coding Platform

[GitHub](#)

Technologies: ASP.NET Core, C#, PostgreSQL, EF Core, Ollama, Roslyn, SignalR

- Built a local-first autonomous software-engineering agent platform with persistent sessions, model-driven JSON/DAG task planning, dependency validation, fan-out/fan-in execution, retries, iteration/repair budgets, approval gates and verification loops.
- Implemented SHA-256 incremental repository indexing, Roslyn symbol extraction, cross-file dependency/reference graphs, context selection and layered short-term/working/long-term memory with semantic, lexical, recency and importance-based retrieval.

ATLAS – Enterprise Intelligent API & Event Platform

[GitHub](#)

Technologies: ASP.NET Core, C#, PostgreSQL, Redis, Kafka, OpenTelemetry

- Engineered a modular enterprise platform combining API management, service registry, traffic routing, Kafka event processing, retries, DLQ/replay, Redis Lua rate limiting, circuit breakers, tenant-aware authorization and operational policy automation.
- Implemented OpenTelemetry-based SLO/error-budget analysis, service health probing, incident state management, deployment-regression analysis, audit/security controls and evidence-based AI operations tooling.

AURORA – Organizational Resilience & Decision Intelligence Architecture

[GitHub](#)

Technologies: ASP.NET Core MVC, C#, PostgreSQL, EF Core, SignalR

- Developed an enterprise digital-twin platform modeling suppliers, factories, products, customers and technology assets through graph intelligence; implemented BFS/DFS, Dijkstra, upstream/downstream propagation, SPOF and bottleneck scoring with deterministic day-by-day disruption simulation.
- Added weighted risk scoring, evidence-backed AI investigation, human-in-the-loop decision approval, audit trails, RBAC, multi-tenancy, SignalR visualization and scenario analysis while explicitly separating implemented functionality from planned optimization, forecasting, RAG and multi-agent features.

SENTINEL – Enterprise Operational Risk & Resilience Simulator

[GitHub](#)

Technologies: ASP.NET Core MVC, C#, PostgreSQL, Redis

- Built an operational-risk simulation platform using dependency graphs, BFS/DFS traversal, single-point-of-failure detection, critical-path analysis, weighted failure propagation and business-impact modeling for revenue, operations, customers and SLAs.
- Implemented genuine Monte Carlo simulation using normal/log-normal sampling and Bernoulli occurrence models, recovery-strategy weighted constrained scoring, resilience metrics, risk heatmaps, background jobs, interactive graph visualization, REST APIs and automated tests.

Aegis – Intelligent Incident Response & Digital Operations Platform

[GitHub](#)

Technologies: ASP.NET Core MVC, C#, PostgreSQL, Redis

- Developed an SRE/AIOps platform with a synthetic infrastructure digital twin and dependency graph implementing the Failure → Detection → Correlation → Root Cause → Impact → Simulation → Remediation → Recovery pipeline.
- Implemented fixed-threshold and rolling z-score anomaly detection, graph-based incident correlation, weighted root-cause ranking using anomaly magnitude/upstream position/deployment/signal evidence, synthetic business-impact analysis and human-approved remediation/runbooks.

NIRBHOR – Bilingual AI Personal Administrative Agent

[GitHub](#)

Technologies: ASP.NET Core MVC, C#, PostgreSQL, EF Core, Identity, SignalR, Anthropic

- Built a bilingual AI administrative agent supporting English, Bangla, Banglish and mixed-language workflows for administrative tasks such as NID correction and passport renewal through conversation, language understanding, agent orchestration, knowledge retrieval and tool execution.
- Implemented versioned prompt management, multilingual/adversarial evaluation datasets, repeatable accuracy-gated evaluation, audit logging, notifications, cost/token tracking, OpenTelemetry and secure Identity workflows while keeping government submission explicitly simulated.

Responsive E-Commerce – ASP.NET MVC/Razor Web Application

[GitHub](#)

Technologies: ASP.NET MVC, C#, Razor Views, Entity Framework, SQL Server

- Developed a responsive database-driven e-commerce application using ASP.NET MVC and Razor with separated Controllers, Models, Services and Views, supporting product management and multi-user shopping-cart functionality.

Skills

Programming Languages: C#, Python, JavaScript.

Web Technologies: ASP.NET Core MVC, ASP.NET MVC, Razor Views, HTML5, CSS3, FastAPI, REST APIs.

Databases: SQL Server, PostgreSQL, MySQL, Entity Framework Core.

Distributed Systems & Infrastructure: RabbitMQ, Redis, Docker, Docker Compose, Background Services, Distributed Locking, Job Queues, Idempotency, Dead-Letter Queues.

Software Architecture: Modular Monolith, Layered Architecture, Domain-Driven Design concepts, Application Services, Dependency Injection, RESTful APIs.

Algorithms & Engineering: BFS, DFS, Dependency Graphs, Failure Propagation, Monte Carlo Simulation, Risk Scoring, Root-Cause Analysis, Optimization, Caching, Query Optimization.

AI & Machine Learning: YOLO, Vision Transformers (ViT), DINOv2, Swin V2, MaxViT, CNN, Random Forest, HOG + SVM, Supervised Contrastive Learning + Classification.

Testing & Tools: xUnit, GitHub, Visual Studio, Google Sheets, Google Data Studio, Retool, Microsoft 365.

Certificates

- **Foundational C# Developer Certification** – Microsoft & freeCodeCamp (2026)
- **Understand ASP.NET Core Fundamentals** – Microsoft Learn (2026)
- **CSS Essentials Certification** – Cisco Networking Academy (2026)
- **Python Essentials 2 (Advanced Python)** – Cisco Networking Academy (2026)
- **JavaScript Essentials 1** – Cisco Networking Academy (2026)
- **English for IT 2** – Cisco Networking Academy | B2 Professional (2026)
- **Legacy Responsive Web Design V8** – freeCodeCamp (2026)
- **HubSpot SEO II Certified** – HubSpot Academy (2026)

Education

B.Sc. in Computer Science – BRAC University **2026**
Dhaka, Bangladesh

Higher Secondary Certificate – Dhaka Residential Model College **2020**
Dhaka, Bangladesh

Secondary School Certificate – Ibn Taimiya School and College **2018**
Dhaka, Bangladesh

Languages

English – Professional Working Proficiency | **Bengali** – Native Speaker